



Sneaker Shoe Assembly

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This document describes the methodology used to construct the Assembly Footwear Datasets, focused exclusively on the assembly phase of footwear, using a modular approach by process and by chemical families. It covers assembly process inputs and outputs only and does **not** include physical footwear materials or components, which are modelled separately. The database was developed based on primary data from a supplier in Vietnam, collected through an onsite visit followed by additional data collection via email.

1. Scope and Functional Unit

1.1 System Analysed

The database covers the following assembly processes:

- Lamination
- Cutting
- HF Welding Process
- Screen Printing
- Insole

- Nosew
- Stitching
- Stock fit
- Assembly

Each of these processes was modelled as an autonomous module, with their respective inputs and outputs.

Physical footwear materials and components are outside the scope of this database and are modelled separately.

1.2 Functional Unit

The functional unit adopted is:

- **1 pair of footwear** ("per pair").

The processes were implemented in **Activity Browser** with **reference product** and **amount = 1 unit**, assuming that "unit" represents the **operation equivalent per pair**, consistent with the structure of the model.

2. Modelling Strategy

2.1 Modular logic

The LCI construction followed a layered logic:

1. Primary chemicals per process

(e.g., base adhesives, resins, hardeners, main solvents)

2. Derived/specific chemicals

(hardener, glue, treatment agents, etc.)

3. Cross-cutting inputs and outputs

- energy
- water
- packaging

- NMVOC
- waste outputs

4. **Modular integration into the final process activities**

(1–9)

This separation enables:

- a clear interpretation of **what enters, where, and why**,
- traceability of **proxy decisions by chemical family**,
- reduced duplication and easier future substitutions

3. Cross-Process Inventory

3.1 Electricity

Electricity supply is modelled in two scenarios:

- **VN**: market for electricity, medium voltage, Vietnam
- **GLO**: market group for electricity, medium voltage, GLO (global average)

Vietnam was selected as the primary scenario because it is the origin of the available primary data. The GLO scenario provides a location-independent reference. Both are applied consistently across all processes involving electricity consumption.

3.2 Water

Across all processes, water was modelled as:

- **water, decarbonised**

This choice was made to maintain internal consistency and reduce methodological noise until factory- or site-specific industrial water data are available.

3.3 NMVOC emissions

Solvent-related emissions were modelled using:

- **non-methane volatile organic compounds (NMVOC)**

- **compartment: air**
- **Biosphere3**

This approach was kept uniform, assuming emissions associated with solvent-based systems in relevant processes.

3.4 Physical footwear materials and components

No physical materials/components of the footwear were included in this database (upper, outsole, foam, knit upper, etc.). The **Assembly Footwear DB** is a **process-only** database covering chemicals, energy, water, packaging, emissions and waste. Material flows are modelled separately (see Section 10).

4. Construction of chemical inputs by phase

4.1 Data source

Chemical quantities per process were calculated based on internal data organised in the Excel file:

- **“01-Material used for Cloud 6”**

This file contains the list and/or percentage shares of chemicals used by formulation type and by process.

4.2 Calculation method applied

The calculation logic was:

- Identify in Excel the **percentage composition per chemical**, according to:
 - formulation type (hardener, glue, treatment agents, etc.)
 - and associated process (lamination, stock fit, screen printing, etc.)
- Define the **total formulation mass per pair**
(or the internally defined reference mass for each process).
- Convert percentages into absolute quantities by applying:

$$m_i = m_{total} \times \%_i$$

Where:

- m_i = mass of chemical i per pair
- m_{total} = total formulation mass associated with the process
- $\%_i$ = chemical percentage in the Excel dataset
- Add the final values to the inventory for the corresponding process module.

This procedure ensures that:

- quantities reflect the **real internal composition structure**,
- and the model does not depend solely on generic defaults.

5. Proxies and process-specific decisions

5.1 Lamination

Within lamination chemicals, the internal decision was maintained to consider:

- **Vinyl Acetate – Ethylene copolymers**
- and **Vinyl Acetate**

as represented under the same **vinyl acetate** proxy family for operational consistency within the current dataset version.

5.2 Screen Printing

Proxies were used for resin families, pigments, fillers and solvents, including:

- **polyurethane (flexible foam)** as a resin proxy
- **titanium dioxide + carbon black (50/50)** for pigments
- **brass** for metallic powders
- **silica sand** for texture powders
- **flat glass** for reflective compounds
- **LDPE fine** for matting agent

- **propylene glycol** as humectant
- solvents such as **ethyl acetate** and **n-butyl acetate**

Selection was guided by functional equivalence and compatibility with availableecoinvent entries.

5.3 Stitching

The following proxies were used:

- **SBR rubber** → proxy for synthetic rubber
- **Terpene resin** → proxied by **rosin size**

representing auxiliary materials typically associated with stitching/assembly chemistry.

5.4 Stock fit

Stock fit includes the most complex chemical set and was modelled using carefully selected proxies, including:

- **1-Ethylpyrrolidin-2-one (NEP)** → proxy **NMP**
- **UV-curable resin (urethane acrylate system)** modelled as a representative mixture:
 - urethane acrylate oligomer → proxy PU
 - reactive acrylate diluents → proxy butyl acrylate
 - photoinitiator → chemical, organic
 - plus additional electricity for UV curing

Additionally:

- Methoxypropyl acetate 2 → **DPGME**
- Tetramethylene dimethacrylate → **methyl methacrylate**
- 2-Hydroxyethyl methacrylate → **butyl acrylate**
- Phenol, 4-isocyanato-, phosphorothioate (3:1) → **MDI**

5.5 Assembly (final process)

In the final assembly module, solvent and aromatic acid proxies were applied using functional equivalence logic, including:

- 1,3-dimethylimidazolidin-2-one → **DMF**
- m-toluic acid → **benzoic acid**
- N,N-diethylformamide → **DMF**
- Dimethyl glutarate → **dimethyl malonate**

Internal cut-off

The chemical **SW-08AT** was not included in the assembly process because it is:

- **< 1% (0.10%)**

therefore below the internally defined relevance threshold.

6. Results (EF v3.1 — Climate Change)

Two electricity scenarios were modelled and compared using **ecoinvent 3.12** (cut-off):

- **VN:** market for electricity, medium voltage, Vietnam
- **GLO:** market group for electricity, medium voltage, GLO (global average)

GWP100 results per process (per pair):

Rank	Process	GWP100 VN (kg CO ₂ -eq)	GWP100 GLO (kg CO ₂ -eq)	Delta
1	8 - Stock fit	0.658	0.425	-35%
2	9 - Assembly	0.423	0.126	-70%
3	7 - Stitching	0.154	0.149	-3%
4	6 - Nosew	0.151	0.060	-60%
5	1 - Lamination	0.133	0.133	~0%
6	2 - Cutting	0.048	0.065	+36%
7	3 - HF Welding	0.058	0.017	-71%
8	4 - Screen	0.039	0.039	~0%

Rank	Process	GWP100 VN (kg CO ₂ -eq)	GWP100 GLO (kg CO ₂ -eq)	Delta
	Printing			
9	5 - Insole	~0	~0	-
	Total	1.664	1.014	-39%

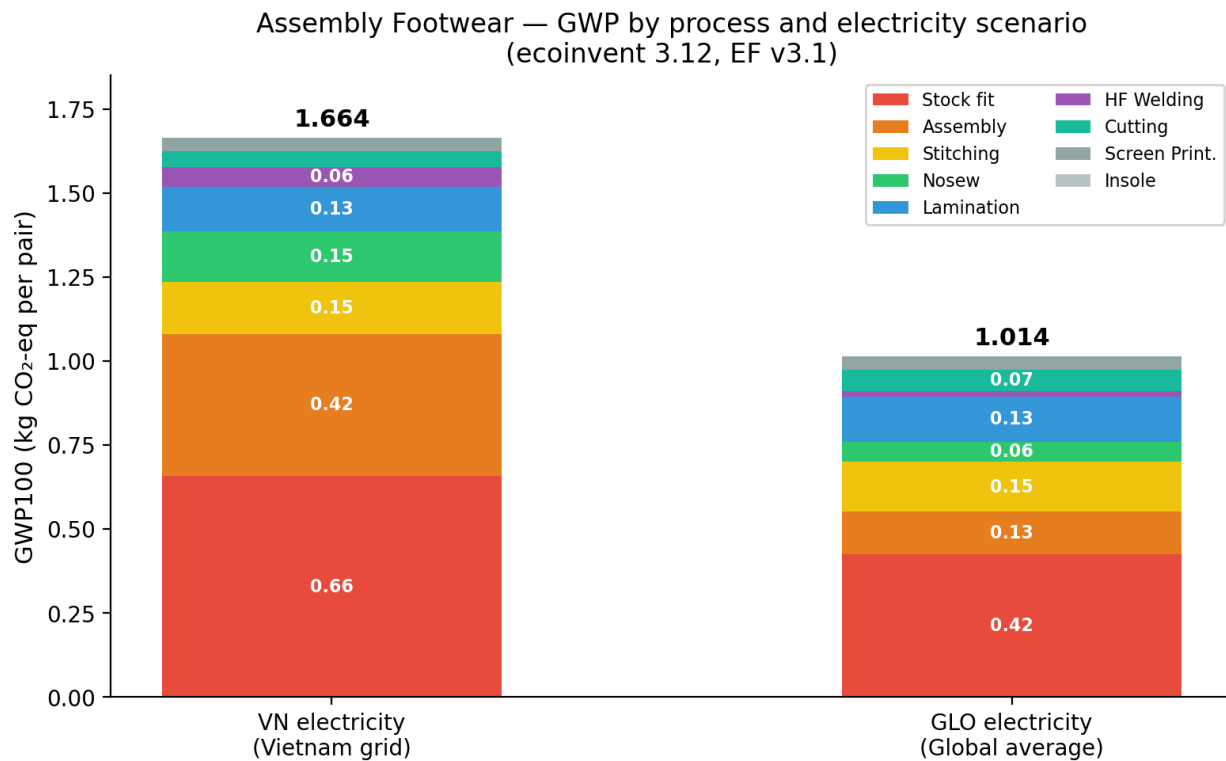


Fig. 1 — GWP100 by process and electricity scenario (ecoinvent 3.12, EF v3.1)

Key findings:

- **Stock fit** is the dominant climate-change driver in both scenarios (~40% of total), driven by chemically intensive bonding/treatment systems and electricity for UV curing.
- Switching from VN to GLO electricity reduces total GWP by **39%** (1.664 → 1.014 kg CO₂-eq).

- Electricity-intensive processes (Assembly –70%, HF Welding –71%, Nosew –60%) are most sensitive to grid choice.
- Chemistry-dominated processes (Lamination, Screen Printing) are virtually unaffected.

7.1 Contribution analysis — Stock fit (VN scenario, 0.658 kg CO₂-eq)

Input	Amount	Unit	GWP (kg CO ₂ -eq)	Share
Electricity (medium voltage, VN)	0.454	kWh	0.233	35.4%
Methyl ethyl ketone	0.028	kg	0.194	29.4%
Ethyl acetate	0.014	kg	0.056	8.5%
Oxalic acid	0.003	kg	0.052	7.9%
Corrugated board box (packaging)	0.027	kg	0.036	5.4%
PU adhesive	0.006	kg	0.030	4.5%
PU flexible foam	0.004	kg	0.024	3.7%
Other (11 inputs)			0.033	5.2%

Stock fit is driven primarily by **electricity** (35%) and **methyl ethyl ketone** (29%). Together these two inputs account for nearly two-thirds of the process footprint. Methyl ethyl ketone (butan-2-one, CAS 78-93-3) has a high emission factor inecoinvent 3.12 (~7.1 kg CO₂-eq/kg for RoW, ~6.3 for RER) — roughly 2× higher than acetone and 1.8× higher than ethyl acetate — due to petrochemical-intensive production from butane oxidation. The largest single consumer is the PL55 treatment agent formulation (17 g/pair).

7.2 Contribution analysis — Stitching (VN scenario, 0.154 kg CO₂-eq)

Input	Amount	Unit	GWP (kg CO ₂ -eq)	Share
Methyl ethyl ketone	0.009	kg	0.058	37.6%
Ethyl acetate	0.013	kg	0.053	34.2%
Rosin size	0.003	kg	0.010	6.3%
Synthetic rubber	0.003	kg	0.009	6.0%
Methylcyclohexane	0.003	kg	0.007	4.9%
LDPE granulate (packaging)	0.002	kg	0.007	4.8%
Electricity (medium voltage, VN)	0.010	kWh	0.005	3.4%
Other (3 inputs)			0.004	2.9%

Stitching is dominated by **solvents** (methyl ethyl ketone + ethyl acetate = 72%), with electricity playing a minor role (3.4%). Methyl ethyl ketone alone contributes 38% despite a small mass (9 g/pair), again due to its high emission factor. This solvent-heavy profile explains why switching to GLO electricity barely affects this process (–3%).

7. Limitations and open items

Item	Status	Next step
Physical footwear materials not included	By design	Material flows modelled separately and coupled later
NMVOC emissions uniform across processes	Simplification	Differentiate by abatement efficiency if data available
Chemical proxies may affect toxicity categories	Known limitation	Refine proxies as more specific datasets become available

8. Record of methodological decisions

- Modular model by process and chemical families

- FU defined **per pair**
- Two electricity scenarios: VN (market for electricity, medium voltage, Vietnam) and GLO (market group for electricity, medium voltage, GLO)
- Water standardised as **water, decarbonised**
- NMVOC emissions to air via **Biosphere3**
- Proxies applied using functional equivalence logic
- Cut-off applied to **SW-08AT (<1%)**
- Background database: **ecoinvent 3.12** (cut-off)

9. Treatment of Material Losses, Input–Output Ratios (IOR) and Defect Rate

9.1 Rationale

In parallel to the process-level assembly database, material efficiency losses occurring along the footwear manufacturing chain were quantified using **Input–Output Ratios (IORs)** derived from primary gross and net material consumption data.

This approach allows us to:

- explicitly capture **cutting losses, trimming losses, yield losses and inefficiencies**,
- separate **process chemistry and energy modelling** from **material flow efficiency**,
- and apply losses in a **transparent, auditable and harmonised manner** across the full product system.

9.2 Definition of Input–Output Ratio (IOR)

For each material flow, the Input–Output Ratio was calculated as:

$$\text{IOR} = \frac{\text{Gross material input}}{\text{Net material incorporated into product}}$$

Example:

- Gross material = 105 g
- Net material = 100 g
- IOR = 1.05

An IOR equal to 1 indicates no loss, while values above 1 reflect material losses upstream of the final product.

9.3 Data source and calculation

IORs were computed using the **Materials / Raw Materials table**, based on primary client data, including:

- material specifications,
- gross input quantities per process,
- net quantities effectively incorporated into footwear components.

For each process (lamination, cutting, HF welding, stitching, stock fit, assembly, etc.), individual IORs were calculated at material level.

9.4 Modelling approach and application point

To preserve model clarity and avoid double counting, the following modelling strategy was adopted:

- **All process modules** (chemicals, energy, water, NMVOC, waste) are modelled **per functional unit (1 pair)** assuming ideal yield.
- **Material losses (IORs)** are **not applied within individual process modules**.
- Instead, an **average material IOR**, aggregated across the full process chain, is applied **once at the end of the assembly chain**, on top of the net material demand.

This ensures that:

- chemical and energy intensities remain comparable across scenarios,
- material inefficiencies are accounted for consistently,
- and future updates to loss rates do not require re-building process inventories.

9.5 Defect rate treatment

In addition to process-related material losses, a **defect rate** (rejected or non-conforming pairs) is applied **on top of the average IOR**, reflecting manufacturing quality losses.

The total material correction factor is therefore:

$$\text{Total factor} = \text{Average IOR} \times (1 + \text{Defect rate})$$

This defect rate is applied at the **final product level**, consistent with industrial practice and avoiding distortion of upstream process intensities.

Defect Rate: 0.2%

9.6 Assumptions and limitations

- IORs are derived from available primary data and may evolve as additional materials or production lines are integrated.
- Averages are used to balance representativeness and model stability.
- Losses are applied to **physical material flows only**, not to chemicals, energy or water used in assembly processes.
- The approach is conservative and aligned with LCA best practice for complex manufactured products

10. Reference flow and scope

Reference flow: Assembly Footwear - Process | Functional unit: **1 pair**

This value represents **assembly operations only**, including chemicals, electricity, water, packaging, NMVOC emissions, and process-related waste — **excluding** physical footwear materials and components, which are modelled in a separate system block.

The user is responsible for including material inputs and accounting for material losses (IOR) and defect rates as documented in Section 9.